

Solution: Possible. This means your computer is five times faster than my computer.

88) Your computer is 250% slower than mine.

89) **Average Percentages.** You are a teacher. Your first-period class, with 25 students, had a mean score of 86% on the midterm exam. Your second-period class, with 30 students, had a mean score of 84% on the same exam. Does it follow that the mean score for both classes combined is 85%? Explain.

Solution: No, the mean score for both classes is not 85%. Assuming a 100-point exam, the first class scored a total of $25 \times 86 = 2150$ points, and the second class scored a total of $30 \times 84 = 2520$ points. Thus the combined number of points scored by 55 students is $2150 + 2520 = 4670$, and the mean score for both classes is $4670/55 = 84.9\%$.

90) **Average Percentages.** A player has a batting average over many games of .400. In his next game, he goes 2 for 4, which is a batting average of .500 for the game. Does it follow that his new batting average is $(.400 + .500)/2 = .450$? Explain.

Unit-M4.2

SIMPLE AND COMPOUND INTEREST

Using and Understanding Mathematics: A Quantitative Reasoning Approach (6th Ed.) by Jeffrey Bennett, William, Briggs

(4B)

Simple interest: is interest paid only on original investment not on any interest on later date.

The Power of Compounding

Simple versus Compound Interest

Imagine that you deposit \$1000 in Honest John's Money Holding Service, which promises to pay 5% interest each year. At the end of the first year, Honest John's sends you a check for

$$5\% \times \$1000 = 0.05 \times \$1000 = \$50$$

Because you receive a check for your interest, your balance with Honest John's remains \$1000, so you get the same \$50 payment at the end of the second year, and the same again for the third year. Your total interest for the three years is

$$3 \times \$50 = \$150$$

Therefore, your original investment of \$1000 grew in value to \$1150. Honest John's method of payment represents **simple interest**, in which interest is paid only on your initial investment. More generally, the amount of money on which interest is paid is called the **principal**.

Now, suppose that you place the \$1000 in a bank account that pays the same 5% interest once a year. But instead of paying you the interest directly, the bank adds the interest to your account. At the end of the first year, the bank deposits $5\% \times \$1000 = \50 interest in to your account, raising your balance to \$1050. At the end of the second year, the bank again pays you 5% interest. This time, however, the 5% interest is paid on the balance of \$1050, so it amounts to

$$5\% \times \$1050 = 0.05 \times \$1050 = \$52.50$$

Adding this \$52.50 raises your balance to $\$1050 + \$52.50 = \$1102.50$

This is the new balance on which your 5% interest is computed at the end of the third year, so your third interest payment is

$$5\% \times \$1102.50 = 0.05 \times \$1102.50 = \$55.13$$

Therefore, your balance at the end of the third year is

$$\$1102.50 + \$55.13 = \$1157.63$$

Despite identical interest rates, you end up with \$7.63 more if you use the bank instead of Honest John's. The difference comes about because the bank pays you *interest on the interest* as well as on the original principal. This type of interest payment is called **compound interest**.

Definitions

The **principal** in financial formulas is the balance upon which interest is paid.

Simple interest is interest paid only on the original investment and not on any interest added at later dates.

Compound interest is interest paid both on the original investment and on all interest that has been added to the original investment.

1- Example- 1: Savings Bond

While banks almost always pay compound interest, bonds usually pay simple interest. Suppose you invest \$1000 in a savings bond that pays simple interest of 10% per year. How much total interest will you receive in 5 years? If the bond paid compound interest, would you receive more or less total interest? Explain.

Solution: simple interest, every year you receive the same interest payment of $10\% \times \$1000 = \100 .

Therefore, you receive a total of \$500 in interest over 5 years. With **compound interest**, you receive more than \$500.

Because the interest each year is calculated on your growing balance rather than on your original investment. For example, because your first interest payment of \$100 raises your balance to \$1100, your next

To find the total balance, we could continue the calculations to 535 years. Fortunately, there's a much easier way. The 4% annual interest rate means that each end-of-year balance is 104% of, or 1.04 times, the previous year's balance. As shown in the last column of Table 4.1, we can get each balance as follows:

- The balance at the end of 1 year is the initial deposit of \$224 times 1.04:

$$\$224 \times 1.04 = \$232.96$$
- The balance at the end of 2 years is the 1-year balance times 1.04, which is equal to the initial deposit times $(1.04)^2$:

$$\$224 \times 1.04 \times 1.04 = \$224 \times (1.04)^2 = \$242.28$$
- The balance at the end of 3 years is the 2-year balance times 1.04, which is equal to the initial deposit times $(1.04)^3$:

$$\$224 \times 1.04 \times 1.04 \times 1.04 = \$224 \times (1.04)^3 = \$251.97$$

Continuing the pattern, we find that the balance after Y years is the initial deposit times 1.04 raised to the Y th power. For example, the balance after $Y = 10$ years is

$$\$224 \times (1.04)^{10} = \$331.57$$

We can generalize this result by looking carefully at the previous equation. Notice that \$224 is the initial deposit, which we will refer to as the **starting principal**. The 1.04 is 1 plus the interest rate of 4%, or 0.04. The exponent 10 is the number of times that the interest has been compounded. Let's write the equation again, adding these identifiers and turning it around to put the result on the left:

$$\underbrace{\$331.57}_{\text{accumulated balance, } A} = \underbrace{\$224}_{\text{starting principal, } P} \times \underbrace{(1.04)}_{1 + \text{interest rate}}^{10} \leftarrow \text{number of compounding periods}$$

When interest is compounded just once a year, as it is in this case, the interest rate is called the **annual percentage rate**, or **APR**. The number of compounding periods is then simply the number of years Y over which the principal earns interest. We therefore obtain the following general formula for interest compounded once a year.

The Compound Interest Formula (for Interest Paid Once a Year)

$$A = P \times (1 + \text{APR})^Y$$

where

- A = Accumulated balance after Y years
- P = Starting principal
- APR = Annual percentage rate (as a decimal)
- Y = Number of years

Notes: (1) The starting principal, P , is often called the **present value (PV)**, because we usually begin a calculation with the amount of money in an account at present. (2) The accumulated balance, A , is often called the **future value (FV)**, because it is the amount that will be accumulated at some time in the future. (3) When using this formula, you must express the APR as a decimal rather than as a percentage.

Technical Note

For the more general case in which the interest rate may not be set on an annual (APR) basis, the compound interest formula is written

$$A = P \times (1 + i)^N$$

Where i is the interest rate and N is the total number of compounding periods.

In the New College case, the annual interest rate is $\text{APR} = 4\% = 0.04$, and interest is paid over a total of 535 years. The accumulated balance after $Y = 535$ years is

$$A = P \times (1 + \text{APR})^Y = \$224 \times (1 + 0.04)^{535} = \$224 \times (1.04)^{535} = \$224 \times 1,296,691,085 \\ \approx \$2.9 \times 10^{11} = \$290 \text{ billion}$$

As the administrator claimed, a 4% interest rate for 535 years would make the original \$224 debt grow to \$290 billion. (And note that in the additional years that have passed since the debt was discovered in 1996, the amount due would have more than doubled again.)

Example-2: Simple and Compound Interest

You invest \$100 in two accounts that each pays an interest rate of 10% per year, but one pays simple interest and the other pays compound interest. Make a table to show the growth of each account over a 5-year period. Use the compound interest formula to verify the result in the table for the compound interest case.

Solution The simple interest is the same absolute amount each year: $10\% \times \$100 = \10 . The compound interest grows from year to year, because it is paid on the accumulated interest as well as on the starting principal. Table 4.2 summarizes the calculations.

To verify the final entry in the table with the compound interest formula, we use a starting principal $P = \$100$ and an annual interest rate $APR = 10\% = 0.10$ with interest paid for $Y = 5$ years. The accumulated balance A is

$$A = P \times (1 + APR)^Y = \$100 \times (1 + 0.1)^5 = \$100 \times (1.1)^5 = \$100 \times 1.6105 = \$161.05$$

This result agrees with the one in the table. Overall, the account paying compound interest builds to \$161.05 while the simple interest account reaches only \$150, even though both pay at the same 10% rate. Although the 10% interest rate that we assumed here is quite high compared to what most banks pay, the basic point should be clear: For the same interest rate, compound interest is always better for the investor than simple interest.

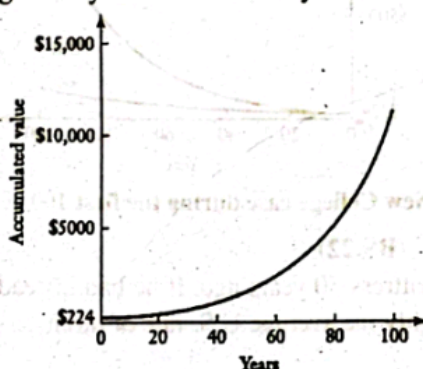
(Same as Q. 55–56)

Table 4.2 Calculations for Example 2 (starting principal $P = \$100$, annual interest rate $APR = 10\%$)

End of Year	Simple Interest Account		Compound Interest Account		
	Interest Paid	Old Balance + Interest = New Balance	Interest Paid	Old Balance + Interest = New Balance	
1	$10\% \times \$100 = \10	$\$100 + \$10 = \$110$	$10\% \times \$100 = \10	$\$100$	$+ \$10 = \110
2	$10\% \times \$100 = \10	$\$110 + \$10 = \$120$	$10\% \times \$110 = \11	$\$110$	$+ \$11 = \121
3	$10\% \times \$100 = \10	$\$120 + \$10 = \$130$	$10\% \times \$121 = \12.10	$\$121$	$+ \$12.10 = \133.10
4	$10\% \times \$100 = \10	$\$130 + \$10 = \$140$	$10\% \times \$133.10 = \13.31	$\$133.10$	$+ \$13.31 = \146.41
5	$10\% \times \$100 = \10	$\$140 + \$10 = \$150$	$10\% \times \$146.41 = \14.64	$\$146.41$	$+ \$14.64 = \161.05

Compound Interest as exponential Growth

The New College case demonstrates the remarkable way in which money can grow with compound interest. Fig. shows how the value of the New College debt rises during the first 100 years, assuming a starting value of \$224 and an interest rate of 4% per year. Note that while the value rises slowly at first, it rapidly accelerates, so in later years the value grows by much more each year than it did during earlier years.



The value of the debt in the New College case during the first 100 years, at an interest rate of 4% per year. Note that the value rises much more rapidly in later years than in earlier years—a hallmark of exponential growth.

This rapid growth is a hallmark of what we generally call *exponential growth*. You can see how exponential growth gets its name by looking again at the general compound interest formula:

$$A = P \times (1 + APR)^Y$$

Because the starting principal P and the interest rate APR have fixed values for any particular compound interest calculation, the growth of the accumulated value A depends only on Y (the number of times interest has been paid), which appears in the *exponent* of the calculation.

Exponential growth is one of the most important topics in mathematics, with applications that include population growth, resource depletion, and radioactivity.

Example- 3: New College Debt at 2%

If the interest rate is 2%, calculate the amount due to New College using

- a. simple interest b. compound interest

Solution

a) The following steps show the simple interest rate calculation for a starting principal

$P = \$224$ and an annual interest rate of 2%:

1) The simple interest due each year is 2% of the starting principal: $2\% \times \$224 = 0.02 \times \$224 = \$4.48$

- 2) Over 535 years, the total interest due is: $535 \times \$4.48 = \2396.80
 3) The total due after 535 years is the starting principal plus the interest: $\$224 + \$2396.80 = \$2620.80$
 With simple interest, the payoff amount after 535 years is $\$2620.80$.

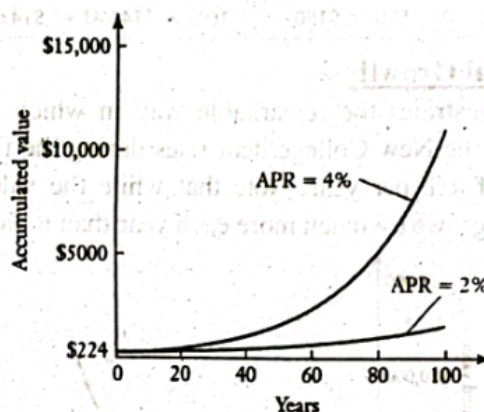
b) To find the amount due with compound interest, we set the annual interest rate to $APR = 2\% = 0.02$ and the number of years to $Y = 535$. Then we use the formula for compound interest paid once a year:

$$A = P \times (1 + APR)^Y = \$224 \times (1 + 0.02)^{535} = \$224 \times (1.02)^{535} = \$224 \times 39,911 = \$8.94 \times 10^6$$

The amount due with compound interest is about \$8.94 million—far higher than the amount due with simple interest. (Same as Q. 57–58)

Effects of Interest Rate Changes

Notice the remarkable effects of small changes in the compound interest rate. In Example 3, we found that a 2% compound interest rate leads to a payoff amount of \$8.94 million after 535 years. Earlier, we found that a 4% interest rate for the same 535 years leads to a payoff amount of \$290 billion—which is more than 30,000 times as large as \$8.94 million. Fig contrasts the values of the New College debt during the first 100 years at interest rates of 2% and 4%. Note that the rate change doesn't make much difference for the first few years, but over time the higher rate becomes far more valuable.



This figure contrasts the debt in the New College case during the first 100 years at interest rates of 2% and 4%.

Example- 4: Mattress Investments (BS.22)

Your grandfather put \$100 under his mattress 50 years ago. If he had instead invested it in a bank account paying 3.5% interest compounded yearly (roughly the average U.S. rate of inflation during that period), how much would it be worth now?

Solution The starting principal is $P = \$100$. The annual percentage rate is $APR = 3.5\% = 0.035$. The number of years is $Y = 50$. So the accumulated balance is

$$A = P \times (1 + APR)^Y = \$100 \times (1 + 0.035)^{50} = \$100 \times (1.035)^{50} = \$558.49$$

Invested at a rate of 3.5%, the \$100 would be worth over \$550 today. Unfortunately, the \$100 was put under a mattress, so it still has a face value of only \$100.

Same as Q. 59–62.

Compound Interest Paid More Than Once a Year

Suppose you could put \$1000 into an investment that pays compound interest at an annual percentage rate of $APR = 8\%$. If the interest is paid all at once at the end of a year, you'll receive interest of

$$8\% \times \$1000 = 0.08 \times \$1000 = \$80$$

Therefore, your year-end balance will be $\$1000 + \$80 = \$1080$.

Now, assume instead that the investment pays interest *quarterly*, or four times a year (once every 3 months). The quarterly interest rate is one-fourth of the annual interest rate:

$$\text{Quarterly interest rate} = \frac{APR}{4} = \frac{8\%}{4} = 2\% = 0.02$$

Table 4.3 shows how quarterly compounding affects the \$1000 starting principal during the first year.

27-50: Algebra Review. Use the skills covered in the Brief Review on to solve the following equations.

- 27) $x-3=9$ 28) $y+4=7$ 29) $z-10=6$ 30) $2x=8$
 31) $3p=12$ 32) $4y+2=18$ 33) $5z-1=19$ 34) $1-6y=13$
 35) $3x-4=2x+6$ 36) $5-4s=6s-5$ 37) $3a+4=6+4a$ 38) $3n-16=53$
 39) $6q-20=60+4q$ 40) $5w-5=3w-25$ 41) $t/4+5=25$ 42) $2x/3+4=2x$
 43) $x^2=25$ 44) $y^3=27$ 45) $(x-42)=36$ 46) $p^{1/3}=3$
 47) $(t/3)^2=16$ 48) $w^2+2=27$ 49) $u^9=512$ 50) $v^3+4=68$

51-54: Simple Interest. Calculate the amount of money you will have in the following accounts after 5 years, assuming that you earn simple interest.

✓ 51) You deposit \$700 in an account with an annual interest rate of 4%

Solution: Simple interest for one year = $700 \$ \times 4\% = 28 \$$
 Simple interest for five year = $28 \times 5 \$ = 140 \$$
 Amount of money after five year = $700 + 140 = 840 \$$

52) You deposit \$1200 in an account with an annual interest rate of 3%

Solution: Simple interest for one year = $1200 \$ \times 3\% = 36 \$$
 Simple interest for five year = $36 \times 5 \$ = 180 \$$
 Amount of money after five year = $1200 + 180 = 1380 \$$

53) You deposit \$3200 in an account with an annual interest rate of 3.5%

Solution: Simple interest for one year = $3200 \$ \times 0.035 = 112 \$$
 Simple interest for five year = $112 \times 5 = 560 \$$
 Amount of money after five year = $3200 + 560 = 3760 \$$

54) You deposit \$1800 in an account with an annual interest rate of 3.8%

Solution: Simple interest for one year = $1800 \$ \times 0.038 = 68.4 \$$
 Simple interest for five year = $68.4 \times 5 = 342 \$$
 Amount of money after five year = $1800 + 342 = 2142 \$$

✓ 55-56: Simple versus Compound Interest. Complete the following tables, which show the performance of two investments over a 5-year period. Round all figures to the nearest dollar.

55) Suzanne deposits \$3000 in an account that earns simple interest at an annual rate of 2.5%. Derek deposits \$3000 in an account that earns compound interest at an annual rate of 2.5%.

Year	Suzanne's Annual Interest	Suzanne's Balance	Derek's Annual Interest	Derek's Balance
1				
2				
3				
4				
5				

Solution:

Year	Simple interest		Compound Interest	
	Suzanne's		Derek's	
	Annual interest	Balance	Annual Interest	Balance
1	$3000 \times 2.5\% = 75$	$3000 + 75 = 3075$	$3000 \times 2.5\% = 75$	$3000 + 75 = 3075$
2	$3000 \times 2.5\% = 75$	$3075 + 75 = 3150$	$3075 \times 2.5\% = 77$	$3075 + 77 = 3152$
3	$3000 \times 2.5\% = 75$	$3150 + 75 = 3225$	$3152 \times 2.5\% = 79$	$3152 + 79 = 3231$
4	$3000 \times 2.5\% = 75$	$3225 + 75 = 3300$	$3231 \times 2.5\% = 81$	$3231 + 81 = 3312$
5	$3000 \times 2.5\% = 75$	$3300 + 75 = 3375$	$3312 \times 2.5\% = 83$	$3312 + 83 = 3395$

56) Ariel deposits \$5000 in an account that earns simple interest at an annual rate of 3%. Travis deposits \$5000 in an account that earns compound interest at an annual rate of 3%.

Let's examine what we've found so far for APR = 8%. If interest is compounded annually (once a year), the annual yield is simply APY = APR = 8%. With quarterly compounding, we found APY = 8.243%. With monthly compounding, we found APY = 8.300%. With daily compounding, we found APY = 8.328%. Clearly, more frequent compounding means a higher APY (for a given APR).

TABLE 4.4 Annual Yield (APY) for APR = 8% with Various Numbers of Compounding Periods (n)

n	APY	n	APY
1	8.0000000%	1000	8.3283601%
4	8.2432160%	10,000	8.3286721%
12	8.2999507%	1,000,000	8.3287064%
365	8.3277572%	10,000,000	8.3287067%
500	8.3280135%	1,000,000,000	8.3287068%

However, notice that the change gets smaller as the frequency of compounding increases. For example, changing from annual compounding ($n = 1$) to quarterly compounding ($n = 4$) increases the APY quite a bit, from 8.000% to 8.243%. In contrast, going from monthly ($n = 12$) to daily ($n = 365$) compounding increases the APY only slightly, from 8.300% to 8.328%. In fact, the APY can't get much larger than it already is for daily compounding.

Table 4.4 shows the APY for various compounding periods and Figure 4.4 is a graph of the results. Note that the annual yield does *not* grow indefinitely. Instead, it approaches a *limit* that is very close to the APY of 8.3287068% found for $n = 1$ billion. In other words, even if we could compound *infinitely many times* per year, the annual yield would not exceed 8.3287068%. Compounding infinitely many times per year is called **Continuous Compounding**. It represents the best possible compounding for a particular APR. With continuous compounding, the compound interest formula takes the following special form.

Compound Interest Formula for Continuous Compounding

$$A = P \times e^{(APR \times Y)}$$

Where,

A = accumulated balance after Y years

P = starting principal

APR = annual percentage rate (as a decimal)

Y = number of years

The number e is a special irrational number with a value of $e \approx 2.71828$. You can compute e to a power with the e^x key on your calculator.

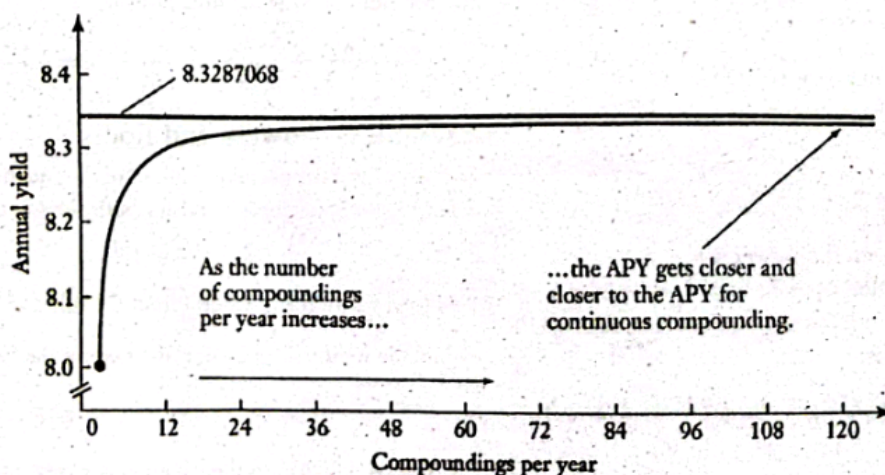


FIGURE 4.4 The annual percentage yield (APY) for APR = 8% depends on the number of times interest is compounded per year.

Example- 7: Continuous Compounding

You deposit \$100 in an account with an APR of 8% and continuous compounding. How much will you have after 10 years?

For interest paid only once each year, we find the balance after 5 years by using the formula for compound interest paid once a year:

$$A = P \times (1 + \text{APR})^Y = \$5000 \times (1 + 0.03)^5 = \$5000 \times (1.03)^5 = \$5796.37$$

After 5 years, monthly compounding gives you a balance of \$5808.08 while annual compounding gives you a balance of \$5796.37. That is, monthly compounding earns \$5808.08 - \$5796.37 = \$11.71 more, even though the APR is the same in both cases. (Same as Q. 63-70)

Annual Percentage Yield (APY)

We've seen that in one year, money grows by *more* than the APR when interest is compounded more than once a year. For example, we found that with quarterly compounding and an 8% APR, a \$1000 principal increases to \$1082.43 in one year. This represents a relative increase of 8.24%:

$$\text{Relative increase} = \frac{\text{Absolute increase}}{\text{Starting principal}} = \frac{\$82.43}{\$1000} = 0.08243 = 8.243\%$$

This relative increase over one year is called the **Annual Percentage Yield (APY)**. Note that it depends only on the annual interest rate (APR) and the number of compounding periods, not on the starting principal.

Definition

The **annual percentage yield (APY)**—also called the *effective yield* or simply the *yield*—is the actual percentage by which a balance increases in one year. It is equal to the APR if interest is compounded annually. It is greater than the APR if interest is compounded more than once a year.

Banks usually list both the annual percentage rate (APR) and the annual percentage yield (APY). However, the APY is what your money really earns and is the more important number when you are comparing interest rates. Banks are required by law to state the APY on interest-bearing accounts.

Example- 6: More Compounding Means a Higher Yield

You deposit \$1000 into an investment with APR = 8%. Find the annual percentage yield with *monthly* compounding and with *daily* compounding.

Solution: The easiest way to find the annual percentage yield is by finding the balance at the end of one year. We have $P = \$1000$, $\text{APR} = 8\% = 0.08$, and $Y = 1$ year. For monthly compounding, we set $n = 12$. At the end of one year, the accumulated balance with monthly compounding is

$$A = P \times \left(1 + \frac{\text{APR}}{n}\right)^{nY} = \$1000 \times \left(1 + \frac{0.08}{12}\right)^{(12 \times 1)} = \$1000 \times (1.006666667)^{12} = \$1083.00$$

Your balance increases by \$83.00, so the annual percentage yield is

$$\text{APY} = \text{relative increase in 1 year} = \frac{\$83.00}{\$1000} = 0.083 = 8.3\%$$

With monthly compounding, the annual percentage yield is 8.3% (which is more than the APR).

Daily compounding means that interest is paid $n = 365$ times per year. At the end of one year, your accumulated balance with daily compounding is

$$A = P \times \left(1 + \frac{\text{APR}}{n}\right)^{nY} = \$1000 \times \left(1 + \frac{0.08}{365}\right)^{(365 \times 1)} = \$1000 \times (1.000219178)^{365} = \$1083.28$$

Your balance increases by \$83.28, so the annual percentage yield is

$$\text{APY} = \text{relative increase in 1 year} = \frac{\text{Absolute increase}}{\text{Starting principal}} = \frac{\$83.28}{\$1000} = 0.08328 = 8.328\%$$

With an APR of 8% and daily compounding, the annual percentage yield is 8.328%, slightly higher than the APY for monthly compounding. (Same as Q. 71-74)

Continuous Compounding

Suppose that interest were compounded more often than daily—say, every second or every trillionth of a second. How would this affect the annual percentage yield?

TABLE 4.3 Quarterly Interest Payments ($P = \$1000$, $APR = 8\%$)

After N Quarters	Interest Paid	New Balance
1st quarter (3 months)	$2\% \times \$1000 = \20	$\$1000 + \$20 = \$1020$
2nd quarter (6 months)	$2\% \times \$1020 = \20.40	$\$1020 + \$20.40 = \$1040.40$
3rd quarter (9 months)	$2\% \times \$1040.40 = \20.81	$\$1040.40 + \$20.81 = \$1061.21$
4th quarter (1 full year)	$2\% \times \$1061.21 = \21.22	$\$1061.21 + \$21.22 = \$1082.43$

Note that the year-end balance with quarterly compounding (\$1082.43) is *greater* than the year-end balance with interest paid all at once (\$1080). That is, when interest is compounded more than once a year, the balance increases by *more* than the APR in 1 year.

We can find the same results with the compound interest formula. Remember that the basic form of the compound interest formula is

$$A = P \times (1 + \text{interest rate})^{\text{number of compoundings}}$$

Where A is the accumulated balance and P is the starting principal. In our current case, the starting principal is $P = \$1000$, the quarterly payments have an interest rate of $APR/4 = 0.02$, and in one year the interest is paid four times. Therefore, the accumulated balance at the end of one year is

$$A = P \times (1 + \text{interest rate})^{\text{number of compoundings}} = \$1000 \times (1 + 0.02)^4 = \$1082.43$$

We see that if interest is paid quarterly, the interest rate at each payment is $APR/4$. Generalizing, if interest is paid n times per year, the interest rate at each payment is APR/n . The total number of times that interest is paid after Y years is nY . We therefore find the following formula for interest paid more than once each year.

Compound Interest Formula for Interest Paid N Times Per Year

$$A = P \left(1 + \frac{APR}{n} \right)^{(nY)}$$

Where,

A = Accumulated balance after Y years

P = Starting principal

APR = Annual percentage rate (as a decimal)

n = Number of compounding periods per year

Y = Number of years

Note that Y is not necessarily an integer; for example, a calculation for six months would have $Y = 0.5$.

Example- 5: Monthly Compounding at 3%

You deposit \$5000 in a bank account that pays an APR of 3% and compounds interest monthly. How much money will you have after 5 years? Compare this amount to the amount you'd have if interest were paid only once each year.

Solution: The starting principal is $P = \$5000$ and the interest rate is $APR = 0.03$. Monthly compounding means that interest is paid $n = 12$ times a year, and we are considering a period of $Y = 5$ years. We put these values into the compound interest formula to find the accumulated balance, A .

$$A = P \times \left(1 + \frac{APR}{n} \right)^{(nY)} = \$5000 \times \left(1 + \frac{0.03}{12} \right)^{(12 \times 5)} = \$5000 \times (1.0025)^{60} = \$5808.08$$

Year	Ariel's Annual Interest	Ariel's Balance	Travis's Annual Interest	Travis's Balance
1				
2				
3				
4				
5				

Solution:

Year	Simple interest		Compound Interest	
	Ariel		Travis	
	Annual interest	Balance	Annual Interest	Balance
1	$5000 \times 3\% = 150$	$5000 + 150 = 5150$	$5000 \times 3\% = 150$	$5000 + 150 = 5150$
2	$5000 \times 3\% = 150$	$5150 + 150 = 5300$	$5150 \times 3\% = 154.5$	$5150 + 154.5 = 5304.5$
3	$5000 \times 3\% = 150$	$5300 + 150 = 5450$	$5304.5 \times 3\% = 159$	$5304 + 159 = 5463$
4	$5000 \times 3\% = 150$	$5450 + 150 = 5600$	$5463 \times 3\% = 163$	$5463 + 163 = 5626$
5	$5000 \times 3\% = 150$	$5600 + 150 = 5750$	$5626 \times 3\% = 168$	$5626 + 168 = 5794$

57–62: Compound Interest. Use the compound interest formula to compute the balance in the following accounts after the stated period of time, assuming interest is compounded annually.

57) \$10,000 is invested at an APR of 4% for 10 years.

Solution: $A = P \times (1 + \text{APR})^Y = \$10,000 \times (1 + 0.04)^{10} = \$10,000 \times (1.04)^{10}$
 $A = \$10,000 \times 1.4802 = \14802.44

58) \$10,000 is invested at an APR of 2.5% for 20 years.

Solution: $A = P \times (1 + \text{APR})^Y = \$10,000 \times (1 + 0.025)^{20} = \$10,000 \times (1.025)^{20}$
 $A = \$10,000 \times 1.6386 = \16386

59) \$15,000 is invested at an APR of 3.2% for 25 years.

Solution: $A = P \times (1 + \text{APR})^Y = \$15,000 \times (1 + 0.032)^{25} = \$15,000 \times (1.032)^{25}$
 $A = \$15,000 \times 2.1978 = \32967.32

60) \$3000 is invested at an APR of 1.8% for 12 years.

Solution: $A = P \times (1 + \text{APR})^Y = \$3000 \times (1 + 0.018)^{12} = \$3000 \times (1.018)^{12}$
 $A = \$3000 \times 1.2387 = \3716.1

61) \$5000 is invested at an APR of 3.1% for 12 years.

Solution: $A = P \times (1 + \text{APR})^Y = \$5000 \times (1 + 0.031)^{12} = \$5000 \times (1.031)^{12}$
 $A = \$5000 \times 1.4424 = \7212.30

62) \$40,000 is invested at an APR of 2.8% for 30 years.

Solution: $A = P \times (1 + \text{APR})^Y = \$40,000 \times (1 + 0.028)^{30} = \$40,000 \times (1.028)^{30}$
 $A = \$40,000 \times 2.2897 = \91588

63–70: Compounding More Than Once a Year. Use the appropriate compound interest formula to compute the balance in the following accounts after the stated period of time.

63. \$10,000 is invested for 10 years with an APR of 2% and quarterly compounding.

Solution: $A = P \times \left(1 + \frac{\text{APR}}{n}\right)^{(nY)}$
 $A = \$10,000 \times \left(1 + \frac{0.02}{4}\right)^{4 \times 10} = \$10,000 \times (1.005)^{40} = \$10,000 \times 1.2207 = 12207.94$

64) \$2000 is invested for 5 years with an APR of 3% and daily compounding.

Solution: $A = P \times \left(1 + \frac{\text{APR}}{n}\right)^{(nY)}$

$$A = \$2000 \times \left(1 + \frac{0.03}{365}\right)^{5 \times 365} = \$200 \times (1.1618) = 2323.6\$$$

65) \$25,000 is invested for 5 years with an APR of 3% and daily compounding.

Solution: $A = P \times \left(1 + \frac{\text{APR}}{n}\right)^{(nY)}$

$$A = \$25000 \times \left(1 + \frac{0.03}{365}\right)^{5 \times 365} = \$200 \times (1.1618) = 29045.68\$$$

66) \$10,000 is invested for 5 years with an APR of 2.75% and monthly compounding.

Solution: $A = P \times \left(1 + \frac{\text{APR}}{n}\right)^{(nY)}$

$$A = \$10,000 \times \left(1 + \frac{0.0275}{12}\right)^{12 \times 5} = \$10,000 \times (1.6022)^{60} = \$10,000 \times 1.14722 = 11472.21$$

67) \$2000 is invested for 15 years with an APR of 5% and monthly compounding.

Solution: $A = P \times \left(1 + \frac{\text{APR}}{n}\right)^{(nY)}$

$$A = \$2000 \times \left(1 + \frac{0.05}{12}\right)^{12 \times 15} = \$2000 \times (1.004)^{180} = \$2000 \times 2.1137 = \$4227.40$$

68) \$30,000 is invested for 15 years with an APR of 4.5% and daily compounding.

Solution: $A = P \times \left(1 + \frac{\text{APR}}{n}\right)^{(nY)}$

$$A = \$30000 \times \left(1 + \frac{0.045}{365}\right)^{15 \times 365} = \$30000 \times (1.9640) = 58918.54\$$$

69. \$25,000 is invested for 30 years with an APR of 3.7% and quarterly compounding.

Solution: $A = P \times \left(1 + \frac{\text{APR}}{n}\right)^{(nY)}$

$$A = \$25000 \times \left(1 + \frac{0.037}{4}\right)^{30 \times 4} = \$25000 \times (3.01892) = 75472.89\$$$

70) \$15,000 is invested for 15 years with an APR of 4.2% and monthly compounding.

Solution: $A = P \times \left(1 + \frac{\text{APR}}{n}\right)^{(nY)}$

$$A = \$15000 \times \left(1 + \frac{0.042}{12}\right)^{15 \times 12} = \$15000 \times (1.0035)^{180} = 15000 \times (1.8755) = \$28133.19$$

71-74: Annual Percentage Yield (APY). Find the annual percentage yield (to the nearest 0.01%) in the following situations.

71) A bank offers an APR of 3.1% compounded daily.

Solution: $\text{APY} = \left[\left(1 + \frac{\text{APR}}{365}\right)^{365} - 1 \right] \times 100\%$

$$\text{APY} = \left[\left(1 + \frac{3.1\%}{365}\right)^{365} - 1 \right] \times 100\% = 3.15\%$$

72) A bank offers an APR of 3.2% compounded monthly.

Solution: $APY = \left[\left(1 + \frac{APR}{12} \right)^{12} - 1 \right] \times 100\%$

$$APY = \left[\left(1 + \frac{3.2\%}{12} \right)^{12} - 1 \right] \times 100\% = 3.25\%$$

73) A bank offers an APR of 1.23% compounded monthly.

Solution: $APY = \left[\left(1 + \frac{APR}{12} \right)^{12} - 1 \right] \times 100\%$

$$APY = \left[\left(1 + \frac{1.23\%}{12} \right)^{12} - 1 \right] \times 100\% = 1.24\%$$

74) A bank offers an APR of 2.25% compounded quarterly.

Solution: $APY = \left[\left(1 + \frac{APR}{4} \right)^4 - 1 \right] \times 100\%$

$$APY = \left[\left(1 + \frac{2.25\%}{4} \right)^4 - 1 \right] \times 100\% = 2.27\%$$

75–80: Continuous Compounding. Use the formula for continuous compounding to compute the balance in the following accounts after 1, 5, and 20 years. Also, find the APY for each account.

75) A \$10,000 deposit in an account with an APR of 3.5%

Solution: Continuous compound = $A = P \times e^{(APR \times Y)}$

$P = 10000$ $APR = 3.5\%$

$Y = 1, \quad A = 10000 \times e^{3.5\% \times 1} = 10,356.20\$$

$Y = 5 \quad A = 10000 \times e^{3.5\% \times 5} = 11,912.46\$$

$Y = 20 \quad A = 10000 \times e^{3.5\% \times 20} = 20,137.53\$$

76) A \$2000 deposit in an account with an APR of 3.1%

Solution: Continuous compound = $A = P \times e^{(APR \times Y)}$

$P = 2000$ $APR = 3.1\%$

$Y = 1 \quad A = 2000 \times e^{3.1\% \times 1} = 2,062.97\$$

$Y = 5 \quad A = 2000 \times e^{3.1\% \times 5} = 2,335.32\$$

$Y = 20 \quad A = 2000 \times e^{3.1\% \times 20} = 3,717.86\$$

77) A \$7000 deposit in an account with an APR of 4.5%

Solution: Continuous compound = $A = P \times e^{(APR \times Y)}$

$P = 7000$ $APR = 4.5\%$

$Y = 1 \quad A = 7000 \times e^{4.5\% \times 1} = 7,322.20\$$

$Y = 5 \quad A = 7000 \times e^{4.5\% \times 5} = 8,766.26\$$

$Y = 20 \quad A = 7000 \times e^{4.5\% \times 20} = 17,217.22\$$

78) A \$3000 deposit in an account with an APR of 7.5%

Solution: Continuous compound = $A = P \times e^{(APR \times Y)}$

$P = 3000$ $APR = 7.5\%$

$Y = 1 \quad A = 3000 \times e^{7.5\% \times 1} = 3,233.65\$$

$Y = 5 \quad A = 3000 \times e^{7.5\% \times 5} = 4,364.97\$$

$Y = 20 \quad A = 3000 \times e^{7.5\% \times 20} = 13,445.07\$$

79) A \$3000 deposit in an account with an APR of 6%

Solution: Continuous compound = $A = P \times e^{(APR \times Y)}$
 $P = 3000$ $APR = 6\%$
 $Y = 1,$ $A = 3000 \times e^{6\% \times 1} = 3185.51\$$
 $Y = 5,$ $A = 3000 \times e^{6\% \times 5} = 4049.58\$$
 $Y = 20,$ $A = 3000 \times e^{6\% \times 20} = 9960.35\$$

80) A \$500 deposit in an account with an APR of 2.7%

Solution: Continuous compound = $A = P \times e^{(APR \times Y)}$
 $P = 500$ $APR = 2.7\%$
 $Y = 1,$ $A = 500 \times e^{2.7\% \times 1} = 513.68\$$
 $Y = 5,$ $A = 500 \times e^{2.7\% \times 5} = 572.27\$$
 $Y = 20,$ $A = 500 \times e^{2.7\% \times 20} = 858.00\$$

81-84: Planning Ahead. How much must you deposit today into the following accounts in order to have \$25,000 in 8 years for a down payment on a house? Assume no additional deposits are made.

81) An account with annual compounding and an APR of 5%

Solution: $A = 25000$ $APR = 0.05$

$$P = \frac{A}{\left(1 + \frac{APR}{n}\right)^{nY}} = \frac{\$25000}{(1 + 0.05)^8} = 16920.98\$$$

82) An account with quarterly compounding and an APR of 4.5%

Solution: $A = 25000$ $APR = 0.045$

$$P = \frac{A}{\left(1 + \frac{APR}{n}\right)^{nY}} = \frac{\$25000}{\left(1 + \frac{0.045}{4}\right)^{8 \times 4}} = 17477.000\$$$

83. An account with monthly compounding and an APR of 6%

Solution: $A = 25000$ $APR = 0.06$

$$P = \frac{A}{\left(1 + \frac{APR}{n}\right)^{nY}} = \frac{\$25000}{\left(1 + \frac{0.06}{12}\right)^{8 \times 12}} = \$15,488.10$$

84) An account with daily compounding and an APR of 4%.

Solution: $A = 25000$ $APR = 4\% = 0.04$

$$P = \frac{A}{\left(1 + \frac{APR}{n}\right)^{nY}} = \frac{\$25000}{\left(1 + \frac{0.04}{365}\right)^{8 \times 365}} = 18154.04\$$$

85-88: College Fund. How much must you deposit today into the following accounts in order to have a \$120,000 college fund in 15 years? Assume no additional deposits are made.

85) An APR of 4%, compounded daily

Solution: $Y = 15$ $A = 120000$ $APR = 0.04$

$$P = \frac{A}{\left(1 + \frac{APR}{n}\right)^{nY}} = \frac{\$120000}{\left(1 + \frac{0.04}{365}\right)^{15 \times 365}} = 56859.56\$$$

86) An APR of 5.5%, compounded daily**Solution:** $Y = 15$ $A = 120000$ $APR = 0.055$

$$P = \frac{A}{\left(1 + \frac{APR}{n}\right)^{nY}} = \frac{\$120000}{\left(1 + \frac{0.055}{365}\right)^{15 \times 365}} = 52591.47\$$$

87) An APR of 2.8%, compounded quarterly**Solution:** $Y = 15$ $A = 120000$ $APR = 0.028$

$$P = \frac{A}{\left(1 + \frac{APR}{n}\right)^{nY}} = \frac{\$120000}{\left(1 + \frac{0.028}{4}\right)^{15 \times 4}} = 78961.07\$$$

88) An APR of 3.5%, compounded monthly**Solution:** $Y = 15$ $A = 120000$ $APR = 0.035$

$$P = \frac{A}{\left(1 + \frac{APR}{n}\right)^{nY}} = \frac{\$120000}{\left(1 + \frac{0.035}{12}\right)^{15 \times 12}} = \$71040.91$$

4C**SAVINGS PLANS AND INVESTMENTS**

Suppose you want to save money, perhaps for retirement or for your child's college expenses. You could deposit a lump sum of money today and let it grow through the power of compound interest. But what if you don't have a large lump sum to start such an account?

For most people, a more realistic way to save is by depositing smaller amounts on a regular basis. For example, you might put \$50 a month into savings. Such long-term **savings plans** are so popular that many have special names—and some get special tax treatment—including Individual Retirement Accounts (IRAs), 401(k) plans, and 529 plans for education.

The Savings Plan Formula

Let's start with an example. Suppose you deposit \$100 into a savings plan at the end of each month. To keep the numbers simple, suppose that your plan pays interest monthly at an annual rate of $APR = 12\%$, or 1% per month.

- You begin with \$0 in the account. At the end of month 1, you make the first deposit of \$100.
- At the end of month 2, you receive the monthly interest on the \$100 already in the account, which is $1\% \times \$100 = \1 . In addition, you make your monthly deposit of \$100. Your balance at the end of month 2 is

$$\underbrace{\$100}_{\text{prior balance}} + \underbrace{\$1.00}_{\text{interest}} + \underbrace{\$100}_{\text{new deposit}} = \$201.00$$

- At the end of month 3, you receive 1% interest on the \$201 already in the account, or $1\% \times \$201 = \2.01 . Adding your monthly deposit of \$100, you have a balance at the end of month 3 of

$$\underbrace{\$201.00}_{\text{prior balance}} + \underbrace{\$2.01}_{\text{interest}} + \underbrace{\$100}_{\text{new deposit}} = \$303.01$$

Table 4.5 continues these calculations through 6 months. In principle, we could extend this table indefinitely—but it would take a lot of work. Fortunately, there's a much easier way: the **savings plan formula** (next page).